

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

INTERPRETATION

$D = 1$: the patient has the disease. $T = 1$: the test result is positive.

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

IDEA

The disease causes the test result. The causal graph has one arrow:



Write the goal as a ratio, and find the numerator and the denominator from the joint distribution:

$$\begin{aligned} p(D \mid T) &= \frac{p(D, T)}{p(T)}, \\ p(D, T) &= p(D) p(T \mid D) && \text{(chain rule; both factors are given),} \\ p(T) &= \sum_d p(D = d, T) && \text{(sum over the parent } D). \end{aligned}$$

SOLUTION

Step 1. Write the joint distribution.

$$p(D, T) = p(D) p(T \mid D) \quad \text{(chain rule; parent of } T: D).$$

Step 2. Write the goal as a ratio.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)}.$$

Step 3. Find the numerator. No sum: all variables are fixed.

$$p(D = 1, T = 1) = p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099.$$

Step 4. Find the denominator. Sum over D :

$$\begin{aligned} p(T = 1) &= \sum_d p(D = d) p(T = 1 \mid D = d) \\ &= p(D = 1) p(T = 1 \mid D = 1) + p(D = 0) p(T = 1 \mid D = 0) \quad (d \in \{0, 1\}) \\ &= 0.000099 + (1 - p(D = 1)) (1 - p(T = 0 \mid D = 0)) \quad \text{(Step 3; complements)} \\ &= 0.000099 + (1 - 0.0001) \cdot (1 - 0.99) \\ &= 0.000099 + 0.9999 \cdot 0.01 \\ &= 0.000099 + 0.009999 = 0.010098. \end{aligned}$$

Step 5. Divide.

$$\begin{aligned}
 p(D = 1 \mid T = 1) &= \frac{0.000099}{0.010098} && \text{(Steps 3, 4)} \\
 &= \frac{99}{10098} && \text{(multiply by } 10^6) \\
 &= \frac{99}{99 \cdot 102} && (10098 = 99 \cdot 102) \\
 &= \frac{1}{102} \approx 0.0098.
 \end{aligned}$$

CONCLUSION

- The probability increases from 0.01% to only 1%.
- In 10 000 people, the test finds $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. Thus most positive results of a mass test for a rare disease are false.
- A second positive independent test, with the prior $1/102$:

$$\begin{aligned}
 p(D = 1 \mid T_1 = 1, T_2 = 1) &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \left(1 - \frac{1}{102}\right) \cdot 0.01} && \text{(Bayes' theorem)} \\
 &= \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} && \left(1 - \frac{1}{102} = \frac{101}{102}\right) \\
 &= \frac{0.99}{0.99 + 101 \cdot 0.01} && \text{(multiply by 102)} \\
 &= \frac{0.99}{0.99 + 1.01} = \frac{0.99}{2} = 0.495.
 \end{aligned}$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

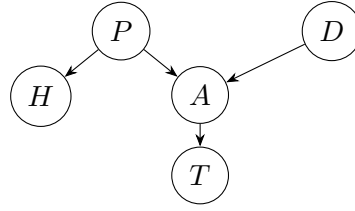


Figure 1: Causal relations for Problem 2

IDEA

Goal: $p(G_x \mid G_y)$ for any groups of variables G_x and G_y . Problem 1 is a special case.

Step 1. Reduce to joint distributions: $p(G_x \mid G_y) = \frac{p(G_x, G_y)}{p(G_y)}$. Now find $p(G)$ for any group G .

Step 2. Find the factors of $p(G)$ with the chain rule. Given: $p(P)$, $p(D)$ for the sources; $p(A \mid P, D)$, $p(H \mid P)$, $p(T \mid A)$ for the other nodes. Only the factors of the sources and arrows with a path to G appear in $p(G)$.

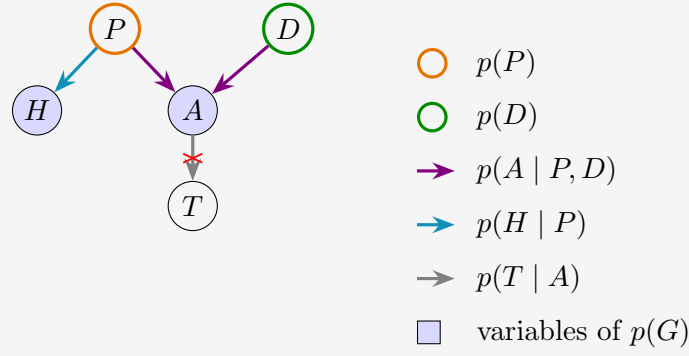
Trick: write the joint distribution of all variables one time. The factor of a node X is $p(X \mid \text{parents of } X)$:

- keep it if X is in G , or if a path goes from X to G ;
- else strike it out.

Example: $G = \{H, A\}$. The chain rule gives

$$\begin{aligned}
 p(P, D, A, H) &= p(P, D, A) p(H \mid P, D, A) && \text{(chain rule)} \\
 &= p(P, D, A) p(H \mid P) && \text{(parent of } H: P) \\
 &= p(P, D) p(A \mid P, D) p(H \mid P) && \text{(chain rule; parents of } A: P, D) \\
 &= p(P) p(D) p(A \mid P, D) p(H \mid P) && \text{(chain rule; } D \text{ has no parents).}
 \end{aligned}$$

From T , no path goes to G , so $p(T \mid A)$ is not in the product:



With the trick: A and H are in G ; $P \rightarrow H$; $D \rightarrow A$; no path goes from T to G :

$$p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) \cancel{p(T | A)}.$$

Step 3. Sum over the kept variables that are not in G . Move the constants out of the sums.

Example: $G = \{H, A\}$.

$$\begin{aligned}
 p(H, A) &= \sum_p \sum_d p(H, A, P = p, D = d) && \text{(sum over } P, D) \\
 &= \sum_p \sum_d p(P = p) p(D = d) && \text{(Step 2)} \\
 &\quad \cdot p(A | P = p, D = d) p(H | P = p).
 \end{aligned}$$

SOLUTION

For each node X , $p(X = 0 | \text{parents}) = 1 - p(X = 1 | \text{parents})$.

Joint distribution.

Chain rule from the children to the parents. Given its parents, a node does not depend on the other nodes.

$$\begin{aligned}
 p(P, D, A, H, T) &= p(P, D, A, H) p(T | P, D, A, H) && \text{(chain rule)} \\
 &= p(P, D, A, H) p(T | A) && \text{(parent of } T: A) \\
 p(P, D, A, H) &= p(P, D, A) p(H | P, D, A) && \text{(chain rule)} \\
 &= p(P, D, A) p(H | P) && \text{(parent of } H: P) \\
 p(P, D, A) &= p(P, D) p(A | P, D) && \text{(chain rule; parents of } A: P, D) \\
 p(P, D) &= p(P) p(D | P) && \text{(chain rule)} \\
 &= p(P) p(D) && (D \text{ has no parents})
 \end{aligned}$$

Thus

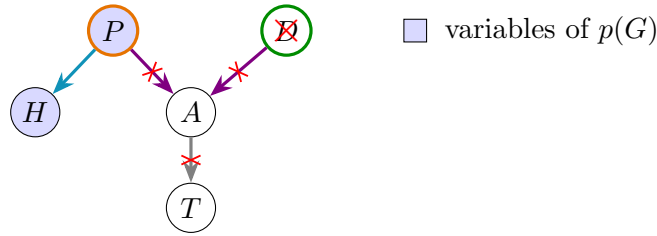
$$p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A).$$

(a) Find $p(P = 1 | H = 1)$.

Step 1 (Idea 1). Write the ratio.

$$p(P = 1 | H = 1) = \frac{p(P = 1, H = 1)}{p(H = 1)}.$$

Step 2 (Idea 2). Numerator: $G = \{P, H\}$.



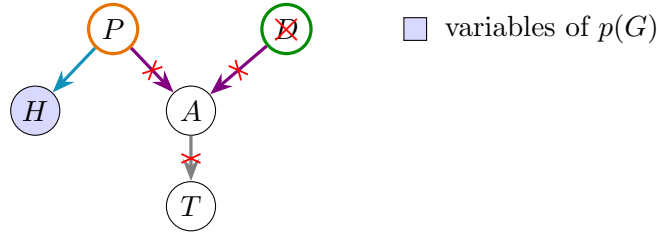
$$p(P, D, A, H, T) = \textcolor{brown}{p(P)} \textcolor{green}{p(D)} \textcolor{red}{p(A | P, D)} \textcolor{blue}{p(H | P)} \textcolor{red}{p(T | A)}$$

$$\implies p(P, H) = \textcolor{brown}{p(P)} \textcolor{blue}{p(H | P)}.$$

Step 3 (Idea 3). Numerator. No sum: no other variables are kept.

$$p(P = 1, H = 1) = p(P = 1) p(H = 1 | P = 1) = 0.2 \cdot 0.9 = 0.18.$$

Step 4 (Idea 2). Denominator: $G = \{H\}$.



$$p(P, D, A, H, T) = \textcolor{brown}{p(P)} \textcolor{green}{p(D)} \textcolor{red}{p(A | P, D)} \textcolor{blue}{p(H | P)} \textcolor{red}{p(T | A)}$$

$$\implies p(P, H) = \textcolor{brown}{p(P)} \textcolor{blue}{p(H | P)}.$$

Step 5 (Idea 3). Denominator.

$$p(H = 1) = \sum_p p(P = p) p(H = 1 | P = p) \quad (\text{Step 4})$$

$$= 0.2 \cdot 0.9 + (1 - 0.2) \cdot 0.2 = 0.18 + 0.16 = 0.34.$$

Step 6. Divide.

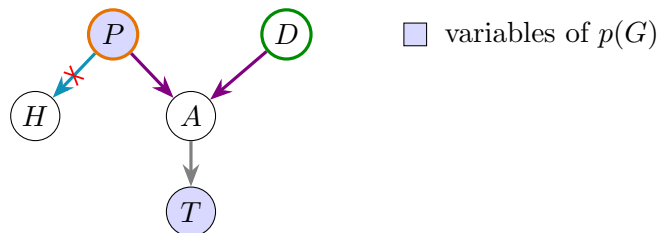
$$p(P = 1 | H = 1) = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

(b) Find $p(P = 1 | T = 1)$.

Step 1 (Idea 1). Write the ratio.

$$p(P = 1 | T = 1) = \frac{p(P = 1, T = 1)}{p(T = 1)}.$$

Step 2 (Idea 2). Numerator: $G = \{P, T\}$.



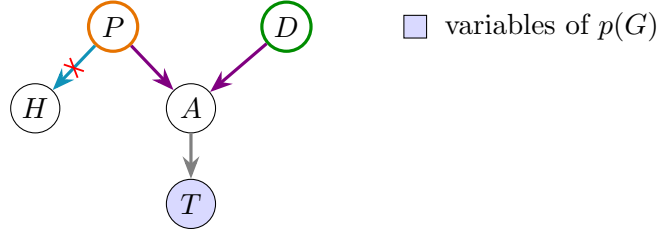
$$p(P, D, A, H, T) = \textcolor{brown}{p}(P) \textcolor{green}{p}(D) \textcolor{violet}{p}(A | P, D) \textcolor{blue}{p}(H | P) p(T | A)$$

$$\implies p(P, D, A, T) = \textcolor{brown}{p}(P) \textcolor{green}{p}(D) \textcolor{violet}{p}(A | P, D) p(T | A).$$

Step 3 (Idea 3). Numerator.

$$\begin{aligned}
& p(P = 1, T = 1) \\
&= \sum_a \sum_d p(P = 1) p(D = d) p(A = a | P = 1, D = d) p(T = 1 | A = a) \quad (\text{Step 2}) \\
&= p(P = 1) \sum_a p(T = 1 | A = a) \sum_d p(A = a | P = 1, D = d) p(D = d) \quad (\text{constants out}) \\
&= 0.2 [0.95 (0.999 \cdot 0.4 + 0.9 \cdot (1 - 0.4)) \quad (a = 1, a = 0) \\
&\quad + 0.5 ((1 - 0.999) \cdot 0.4 + (1 - 0.9) \cdot (1 - 0.4))] \\
&= 0.2 [0.95 \cdot 0.9396 + 0.5 \cdot 0.0604] \\
&= 0.2 [0.89262 + 0.0302] = 0.2 \cdot 0.92282 = 0.184564.
\end{aligned}$$

Step 4 (Idea 2). Denominator: $G = \{T\}$.



$$p(P, D, A, H, T) = \textcolor{brown}{p}(P) \textcolor{green}{p}(D) \textcolor{violet}{p}(A | P, D) \textcolor{blue}{p}(H | P) p(T | A)$$

$$\implies p(P, D, A, T) = \textcolor{brown}{p}(P) \textcolor{green}{p}(D) \textcolor{violet}{p}(A | P, D) p(T | A).$$

Step 5 (Idea 3). Denominator.

$$\begin{aligned}
& p(T = 1) \\
&= \sum_p \sum_a \sum_d p(P = p) p(D = d) p(A = a | P = p, D = d) p(T = 1 | A = a) \quad (\text{Step 4}) \\
&= \sum_p p(P = p) \sum_a p(T = 1 | A = a) \sum_d p(A = a | P = p, D = d) p(D = d) \quad (\text{constants out}) \\
&= 0.184564 + (1 - 0.2) [0.95 (0.9 \cdot 0.4 + 0.01 \cdot 0.6) \quad (p = 1: \text{Step 3}) \\
&\quad + 0.5 ((1 - 0.9) \cdot 0.4 + (1 - 0.01) \cdot 0.6)] \\
&= 0.184564 + 0.8 [0.95 \cdot 0.366 + 0.5 \cdot 0.634] \\
&= 0.184564 + 0.8 [0.3477 + 0.317] = 0.184564 + 0.8 \cdot 0.6647 \\
&= 0.184564 + 0.53176 = 0.716324.
\end{aligned}$$

Step 6. Divide.

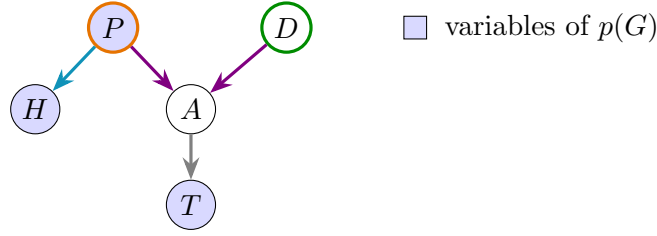
$$p(P = 1 | T = 1) = \frac{0.184564}{0.716324} \approx 0.258.$$

(c) Find $p(P = 1 | T = 1, H = 1)$.

Step 1 (Idea 1). Write the ratio.

$$p(P = 1 | T = 1, H = 1) = \frac{p(P = 1, H = 1, T = 1)}{p(H = 1, T = 1)}.$$

Step 2 (Idea 2). Numerator: $G = \{P, H, T\}$.



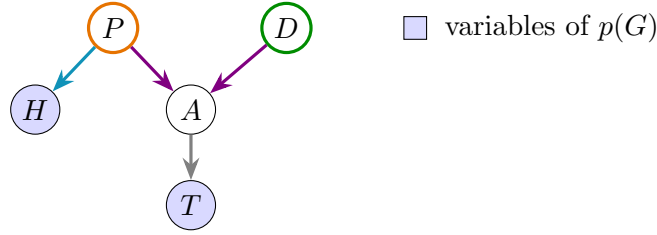
$$p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A)$$

$$\Rightarrow p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A).$$

Step 3 (Idea 3). Numerator.

$$\begin{aligned}
 & p(P = 1, H = 1, T = 1) \\
 &= \sum_a \sum_d p(P = 1) p(D = d) p(A = a | P = 1, D = d) && \text{(Step 2)} \\
 &\quad \cdot p(H = 1 | P = 1) p(T = 1 | A = a) \\
 &= p(H = 1 | P = 1) && \text{(constants out)} \\
 &\quad \cdot p(P = 1) \sum_a p(T = 1 | A = a) \sum_d p(A = a | P = 1, D = d) p(D = d) \\
 &= 0.9 \cdot 0.184564 && \text{((b), Step 3)} \\
 &= 0.1661076.
 \end{aligned}$$

Step 4 (Idea 2). Denominator: $G = \{H, T\}$.



$$p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A)$$

$$\Rightarrow p(P, D, A, H, T) = p(P) p(D) p(A | P, D) p(H | P) p(T | A).$$

Step 5 (Idea 3). Denominator.

$$\begin{aligned}
 & p(H = 1, T = 1) \\
 &= \sum_p \sum_a \sum_d p(P = p) p(D = d) p(A = a | P = p, D = d) && \text{(Step 4)} \\
 &\quad \cdot p(H = 1 | P = p) p(T = 1 | A = a) \\
 &= \sum_p p(H = 1 | P = p) && \text{(constants out)} \\
 &\quad \cdot p(P = p) \sum_a p(T = 1 | A = a) \sum_d p(A = a | P = p, D = d) p(D = d) \\
 &= 0.9 \cdot 0.184564 + 0.2 \cdot 0.53176 && \text{((b), Step 5)} \\
 &= 0.1661076 + 0.106352 = 0.2724596.
 \end{aligned}$$

Step 6. Divide.

$$p(P = 1 | T = 1, H = 1) = \frac{0.1661076}{0.2724596} \approx 0.610.$$

CONCLUSION

$$\underbrace{0.2}_{\text{prior}} \rightarrow \underbrace{0.529}_{H=1}, \underbrace{0.258}_{T=1}, \underbrace{0.610}_{H=1, T=1} .$$

An unhappy teacher is weak evidence: without the party, the teacher is also often unhappy (0.6647 in Step 5 of (b)). Two observations together give the strongest evidence.

3. Let $x_1 \sim \text{Pois}(\lambda_1)$ and $x_2 \sim \text{Pois}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Pois}(\lambda_1 + \lambda_2)$.

IDEA

For $y = f(x_1, \dots, x_n)$ with discrete x_i , split the event $\{y = k\}$ into disjoint events and sum:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n) = \sum_{f(j_1, \dots, j_n) = k} \prod_i p(x_i = j_i) \quad (\text{if independent}).$$

Then simplify the sum and compare it with known distributions.

SOLUTION

Goal: show that $p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^k / k!$ for all $k \geq 0$.

$$\begin{aligned} p(x_1 + x_2 = k) &= p\left(\bigcup_{j=0}^k \{x_1 = j, x_2 = k - j\}\right) && (x_1, x_2 \geq 0) \\ &= \sum_{j=0}^k p(x_1 = j, x_2 = k - j) && (\text{disjoint events}) \\ &= \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j) && (\text{independence}) \\ &= \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!} && (\text{Poisson probabilities}) \\ &= e^{-\lambda_1} e^{-\lambda_2} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (\text{factors without } j \text{ out of the sum}) \\ &= e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && (e^x e^y = e^{x+y}) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} && (\text{multiply and divide by } k!) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} && (\binom{k}{j} = \frac{k!}{j! (k-j)!}) \\ &= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k && (\text{binomial theorem}). \end{aligned}$$

This is $\text{Pois}(\lambda_1 + \lambda_2)$.

CONCLUSION

A sum of independent Poisson variables is a Poisson variable, and the parameters add. By induction, this is true for any finite number of variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

IDEA

$$L(\lambda) = \prod_i p(x_i | \lambda) \rightarrow \ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda) \rightarrow \ell'(\lambda) = 0 \rightarrow \ell''(\lambda) < 0.$$

The logarithm is increasing, so L and ℓ have the same maximum point.

SOLUTION

Let $S = \sum_{i=1}^N x_i$ and $\bar{x} = S/N$.

Step 1. Write the goal.

$$\begin{aligned} \hat{\lambda}_{\text{ML}} &= \arg \max_{\lambda} L(\lambda) \\ &= \arg \max_{\lambda} \ell(\lambda) \end{aligned} \quad (\log \text{ is increasing}).$$

Step 2. Find $\ell(\lambda)$ for Step 1.

$$\begin{aligned} \ell(\lambda) &= \log \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} && (\text{Poisson probabilities}) \\ &= \sum_{i=1}^N \log \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) && (\log \text{ of a product}) \\ &= \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!) && (\log \text{ of each factor}) \\ &= -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i! && (\text{sum each term}). \end{aligned}$$

Step 3. Find the stationary point of ℓ .

$$\begin{aligned} \ell'(\lambda) &= -N + \frac{S}{\lambda}, \\ -N + \frac{S}{\lambda} &= 0 \implies \frac{S}{\lambda} = N \implies \lambda = \frac{S}{N} = \bar{x}. \end{aligned}$$

Step 4. Make sure that it is a maximum.

$$\ell''(\lambda) = -\frac{S}{\lambda^2} < 0 \quad \text{if } S > 0.$$

Thus ℓ is strictly concave, and $\bar{x}$ is the global maximum. If $S = 0$, then $\ell(\lambda) = -N\lambda$ decreases, and the maximum is at $\lambda = 0 = \bar{x}$.

$$\hat{\lambda}_{\text{ML}} = \bar{x}.$$

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

IDEA

The numerator and the denominator of Bayes' theorem contain the same function:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

Do not calculate the integral directly:

$$\begin{aligned} p(X \mid \lambda) p(\lambda) &= C \cdot g(\lambda) && (C \text{ does not depend on } \lambda) \\ g(\lambda) &= Z q(\lambda) && (q \text{ is a known density}) \\ \int CZ q(\lambda) d\lambda &= CZ && (\text{a density integrates to } 1) \\ p(\lambda \mid X) &= \frac{CZ q(\lambda)}{CZ} = q(\lambda). \end{aligned}$$

The same method gives the posterior mean.

SOLUTION

Let $S = \sum_{i=1}^N x_i$, $a' = a + S$, and $b' = b + N$.

Step 1. Write the goal.

$$\hat{\lambda}_B = \mathbb{E}[\lambda \mid X] = \int \lambda p(\lambda \mid X) d\lambda.$$

Step 2. Find $p(\lambda \mid X)$ for Step 1.

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}.$$

Step 2a. Separate the constant in the numerator of Step 2.

$$\begin{aligned} p(X \mid \lambda) p(\lambda) &= \prod_{i=1}^N \left(e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \right) \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\text{likelihood and prior}) \\ &= \frac{\prod_i e^{-\lambda} \cdot \prod_i \lambda^{x_i}}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\text{split the product}) \\ &= \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} && (\prod_i \lambda^{x_i} = \lambda^{\sum_i x_i}) \\ &= \frac{b^a}{\Gamma(a) \prod_i x_i!} \cdot \lambda^S \lambda^{a-1} \cdot e^{-N\lambda} e^{-b\lambda} && (\text{group the factors}) \\ &= \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a'-1} e^{-b'\lambda}}_{g(\lambda)} && (\text{add the exponents}). \end{aligned}$$

Step 2b. Find the known density for g from Step 2a.

$$\mathcal{G}(\lambda \mid a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} = \frac{b'^{a'}}{\Gamma(a')} g(\lambda) \implies g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b').$$

Step 2c. Find the denominator of Step 2.

$$\begin{aligned}
\int Cg(\lambda) d\lambda &= \int C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b') d\lambda && \text{(Step 2b)} \\
&= C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda | a', b') d\lambda && \text{(constants out of the integral)} \\
&= C \frac{\Gamma(a')}{b'^{a'}} && \text{(a density integrates to 1).}
\end{aligned}$$

Step 2d. Substitute Steps 2a–2c into Step 2.

$$\begin{aligned}
p(\lambda | X) &= \frac{Cg(\lambda)}{C \frac{\Gamma(a')}{b'^{a'}}} && \text{(Steps 2a, 2c)} \\
&= \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda | a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} && \text{(Step 2b)} \\
&= \mathcal{G}(\lambda | a', b') && \text{(the constants cancel)} \\
&= \mathcal{G}(\lambda | a + S, b + N).
\end{aligned}$$

Step 3. Substitute Step 2d into Step 1.

$$\begin{aligned}
\mathbb{E}[\lambda | X] &= \int \lambda \mathcal{G}(\lambda | a', b') d\lambda && \text{(Step 2d)} \\
&= \int \lambda \cdot \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} d\lambda && \text{(gamma density)} \\
&= \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda && \text{(constants out of the integral).}
\end{aligned}$$

Step 3a. Find the integral for Step 3 with the same method. Use Step 2b with $a' + 1$ instead of a' :

$$\begin{aligned}
\int \lambda^{a'} e^{-b'\lambda} d\lambda &= \int \lambda^{(a'+1)-1} e^{-b'\lambda} d\lambda && (a' = (a' + 1) - 1) \\
&= \int \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda | a' + 1, b') d\lambda && \text{(Step 2b)} \\
&= \frac{\Gamma(a' + 1)}{b'^{a'+1}} \int \mathcal{G}(\lambda | a' + 1, b') d\lambda && \text{(constants out of the integral)} \\
&= \frac{\Gamma(a' + 1)}{b'^{a'+1}} && \text{(a density integrates to 1).}
\end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$\begin{aligned}
\mathbb{E}[\lambda | X] &= \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} && \text{(Step 3a)} \\
&= \frac{\Gamma(a' + 1)}{\Gamma(a') b'} && (b'^{a'}/b'^{a'+1} = 1/b') \\
&= \frac{a' \Gamma(a')}{\Gamma(a') b'} && (\Gamma(a' + 1) = a' \Gamma(a')) \\
&= \frac{a'}{b'} = \frac{a + S}{b + N}.
\end{aligned}$$

$$\hat{\lambda}_B = \frac{a + S}{b + N}.$$

CONCLUSION

- *Conjugate prior:* gamma prior $\times$ Poisson likelihood $\rightarrow$ gamma posterior. The update only changes the parameters: $a \rightarrow a + S$ (events), $b \rightarrow b + N$ (observations).
- The estimate is a weighted mean of the prior mean and the estimate $\bar{x}$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \bar{x} \xrightarrow{N \rightarrow \infty} \bar{x}.$$

- If $a, b \rightarrow 0$, then $p(\lambda) \propto \lambda^{a-1} e^{-b\lambda} \rightarrow 1/\lambda$, the non-informative prior from the notes below, and $\hat{\lambda}_B \rightarrow S/N = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).